

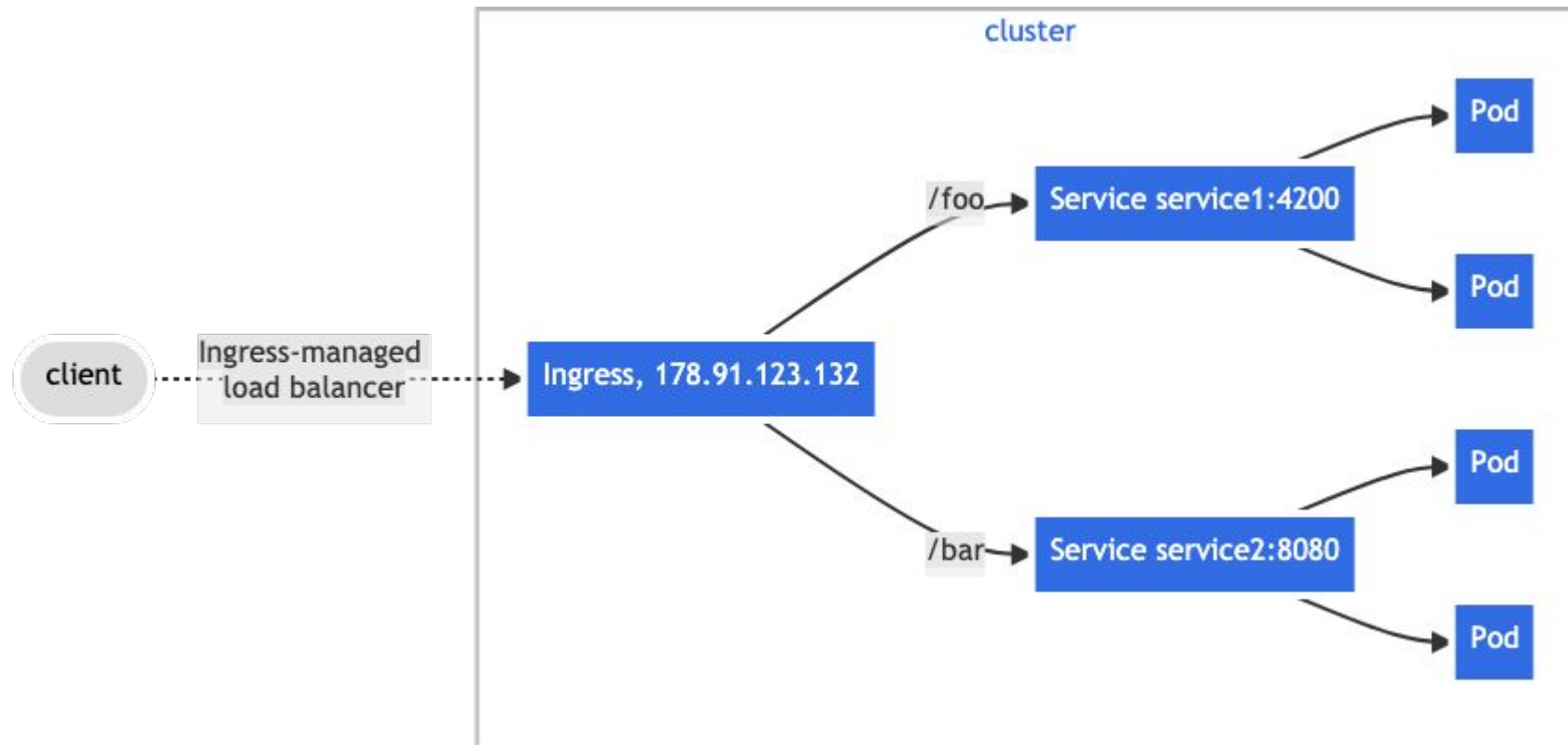


DevOps

Kubernetes Ingress

Kubernetes Ingress

- Kubernetes Ingress is a way to expose HTTP and HTTPS services to the outside world using a single IP address. It is an API object that manages external access to the services in a Kubernetes cluster.



Kubernetes Ingress

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deploy
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx-app
  template:
    metadata:
      labels:
        app: nginx-app
    spec:
      containers:
        - name: nginx-cont
          image: nginx:latest
          ports:
            - containerPort: 80
```

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-svc
spec:
  type: ClusterIP
  selector:
    app: nginx-app
  ports:
    - port: 80
      targetPort: 80
```

Kubernetes Ingress

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: register-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: register-app
  template:
    metadata:
      labels:
        app: register-app
    spec:
      containers:
        - name: register-cont
          image: xeng/spring-register:7
          ports:
            - containerPort: 8080
          volumeMounts:
            - name: register-vol
              mountPath: /src/main/resources/images
      volumes:
        - name: register-vol
          nfs:
            server: 10.148.0.12
            path: /opt/nfs/data
```

```
apiVersion: v1
kind: Service
metadata:
  name: register-svc
spec:
  type: ClusterIP
  selector:
    app: register-app
  ports:
    - port: 8080
```

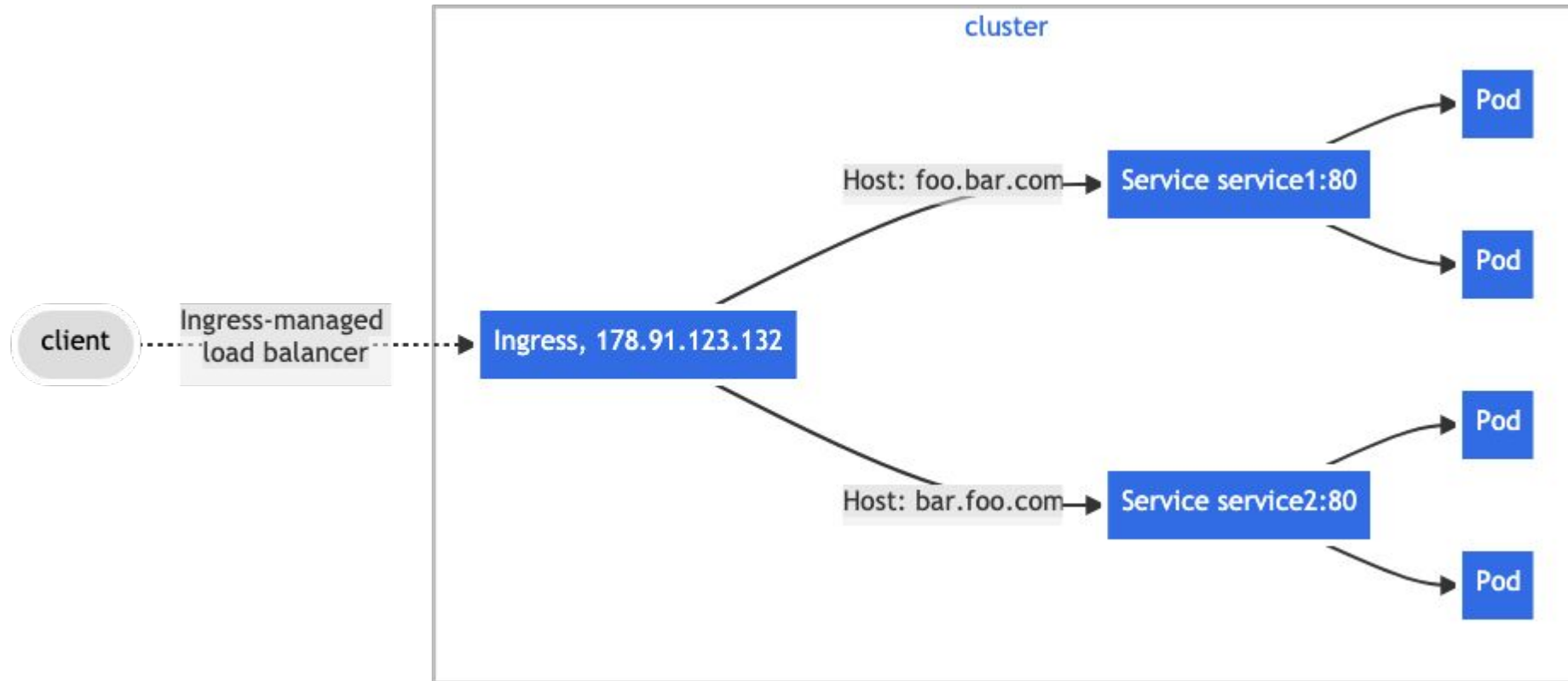
Kubernetes Ingress

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: my-ingress
spec:
  rules:
    - host: register.e-crops.co
      http:
        paths:
          - path: /
            pathType: Prefix
            backend:
              service:
                name: nginx-svc
                port:
                  number: 80
```

Kubernetes Ingress

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: my-ingress2
  annotations:
    nginx.ingress.kubernetes.io/rewrite-target: /$2
spec:
  rules:
    - host: register.e-crops.co
      http:
        paths:
          - path: /spring(/|$)(.*)
            pathType: Prefix
            backend:
              service:
                name: register-svc
                port:
                  number: 8080
```

Kubernetes Ingress



Kubernetes Ingress

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend-deploy
spec:
  replicas: 1
  selector:
    matchLabels:
      app: frontend
  template:
    metadata:
      labels:
        app: frontend
    spec:
      containers:
        - name: frontend-cont
          image: xeng/simple-route:latest
          ports:
            - containerPort: 80
          volumeMounts:
            - name: default-vol
              mountPath: /etc/nginx/conf.d
      volumes:
        - name: default-vol
          configMap:
            name: default-configmap
            items:
              - key: default.conf
                path: default.conf
```

```
apiVersion: v1
kind: Service
metadata:
  name: frontend-svc
spec:
  type: ClusterIP
  selector:
    app: frontend
  ports:
    - port: 80
```

Kubernetes Ingress

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: baseroute-deploy
spec:
  replicas: 1
  selector:
    matchLabels:
      app: baseroute
  template:
    metadata:
      labels:
        app: baseroute
    spec:
      containers:
        - name: baseroute-cont
          image: xeng/base-route:61
          ports:
            - containerPort: 80
          volumeMounts:
            - name: default-vol
              mountPath: /etc/nginx/conf.d
      volumes:
        - name: default-vol
          configMap:
            name: default-configmap
            items:
              - key: default.conf
                path: default.conf
```

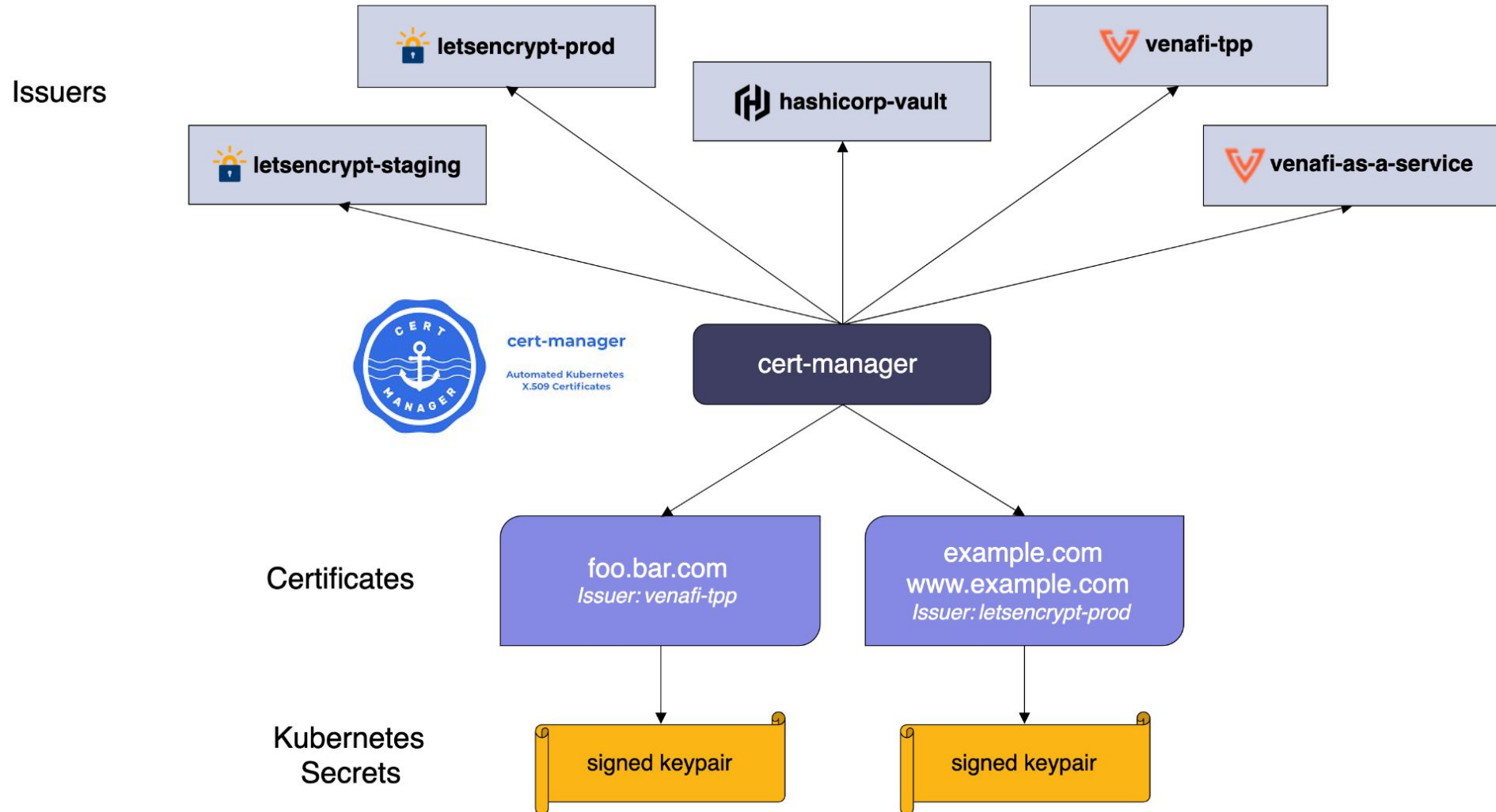
```
apiVersion: v1
kind: Service
metadata:
  name: baseroute-svc
spec:
  type: ClusterIP
  selector:
    app: baseroute
  ports:
    - port: 80
```

Kubernetes Ingress

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: multi-host-ingress
spec:
  rules:
  - host: spring.e-crops.co
    http:
      paths:
      - pathType: Prefix
        path: "/"
        backend:
          service:
            name: register-svc
            port:
              number: 8080
  - host: frontend.e-crops.co
    http:
      paths:
      - pathType: Prefix
        path: "/"
        backend:
          service:
            name: frontend-svc
            port:
              number: 80
```

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: multi-host-ingress2
  annotations:
    nginx.ingress.kubernetes.io/rewrite-target: /$2
spec:
  rules:
  - host: frontend.e-crops.co
    http:
      paths:
      - path: /reactjs(/|$)(.*)
        pathType: Prefix
        backend:
          service:
            name: baseroute-svc
            port:
              number: 80
```

Kubernetes with Cert-Manager



Kubernetes with Cert-Manager

- Cert-manager is an open-source add-on for Kubernetes that automates the management and issuance of TLS certificates. TLS (Transport Layer Security) is a cryptographic protocol that provides secure communication over the internet, and certificates are used to verify the identity of the communicating parties.
- Cert-manager integrates with Let's Encrypt, a free and open Certificate Authority, and other certificate authorities to issue and renew TLS certificates automatically, which makes it easier to secure communication between applications running in Kubernetes clusters. Cert-manager provides a Kubernetes-native way to manage certificates, which means that certificate management can be automated and scaled with the rest of your Kubernetes workloads.

Kubernetes with Cert-Manager

```
kind: ClusterIssuer
metadata:
  name: letsencrypt-prod
  namespace: cert-manager
spec:
  acme:
    # The ACME server URL
    server: https://acme-v02.api.letsencrypt.org/directory
    # Email address used for ACME registration
    email: leangsengk90@gmail.com
    # Name of a secret used to store the ACME account private key
    privateKeySecretRef:
      name: letsencrypt-prod
    # Enable the HTTP-01 challenge provider
    solvers:
      - http01:
          ingress:
            class: nginx
```

Kubernetes with Cert-Manager

```
apiVersion: cert-manager.io/v1
kind: Certificate
metadata:
  name: my-cert
  namespace: default
  annotations:
    kubernetes.io/ingress.class: "nginx"
spec:
  dnsNames:
    - hello.e-crops.co
  secretName: my-tls
issuerRef:
  name: letsencrypt-prod
  kind: ClusterIssuer
```

Kubernetes with Cert-Manager

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: nginx-ingress
spec:
  tls:
    - hosts:
        - hello.e-crops.co
      secretName: my-tls
  rules:
    - host: hello.e-crops.co
      http:
        paths:
          - path: /
            pathType: Prefix
            backend:
              service:
                name: nginx-svc
                port:
                  number: 80
```

Kubernetes with Cert-Manager

Without Certificate

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: nginx-ingress
  annotations:
    cert-manager.io/cluster-issuer: "letsencrypt-prod"
    kubernetes.io/ingress.class: "nginx"
spec:
  tls:
    - hosts:
        - hello.e-crops.co
      secretName: quickstart-example-tls
  rules:
    - host: hello.e-crops.co
      http:
        paths:
          - path: /
            pathType: Prefix
            backend:
              service:
                name: nginx-svc
                port:
                  number: 80
```

Kubernetes with Cert-Manager

The screenshot displays two browser windows. The background window is a React App at `frontend.e-crops.co` with a navigation menu: [To Home](#), [To Profile](#), [To Andrew's Profile](#), and [To Feed](#). The main content area is titled "Home Page!" and features the "KSHRD REGISTRATION" logo with a version "1.0" badge. Below the logo, it states "Online registration platform for entrance exam at KSHRD C" and provides links for "Terms of service", "KSHRD REGISTRATION - Website", "Send email to KSHRD REGISTRATION", and "KSHRD License". At the bottom, there are four controller links: **education-controller** (Education Controller), **generation-controller** (Generation Controller), **home-controller** (Home Controller), and **jwt-authentication-controller** (Jwt Authentication Controller).

The foreground window is the Swagger UI at `spring.e-crops.co/swagger-ui.html`. It shows a "Certificate Viewer: frontend.e-crops.co" modal. The modal has two tabs: "General" (selected) and "Details". The "General" tab displays the following information:

Issued To	
Common Name (CN)	frontend.e-crops.co
Organization (O)	<Not Part Of Certificate>
Organizational Unit (OU)	<Not Part Of Certificate>

Issued By	
Common Name (CN)	R3
Organization (O)	Let's Encrypt
Organizational Unit (OU)	<Not Part Of Certificate>

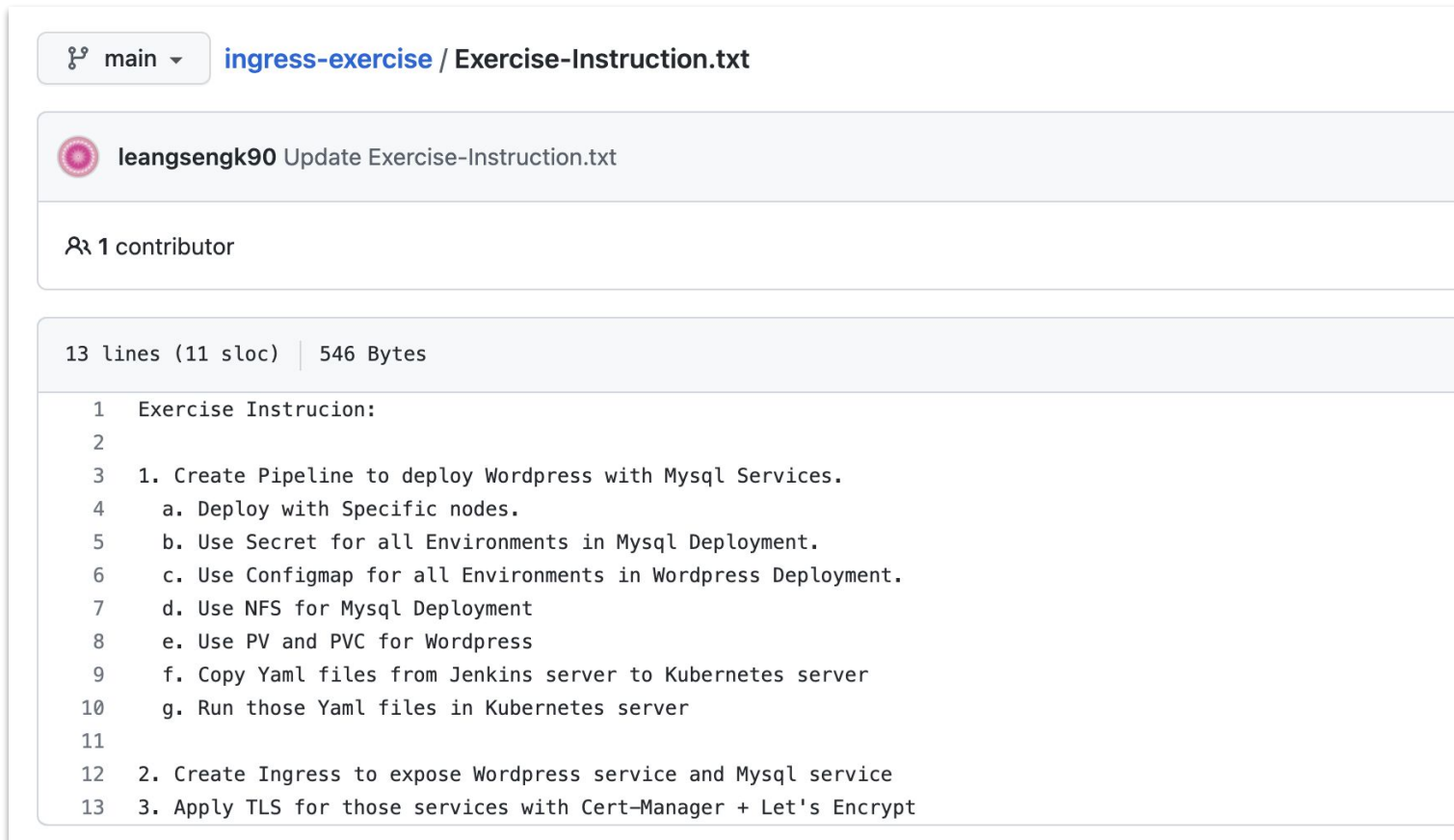
Validity Period	
Issued On	Saturday, March 25, 2023 at 7:04:49 AM
Expires On	Friday, June 23, 2023 at 7:04:48 AM

Fingerprints	
SHA-256 Fingerprint	79 29 8C C4 9E 52 EE 8E AB 21 F3 5B B3 0A B1 63 C1 17 75 8A F5 67 5C 23 A9 3D CB F3 22 47 0A 6D D6 22 8C 09 11 FF F0 84 A0 63 7B 7C 37 B1 26 7E FB 89 30 08
SHA-1 Fingerprint	

Exercise

- Instruction and Resources URL:

<https://github.com/leangsengk90/ingress-exercise.git>



The screenshot shows the GitHub interface for the file 'Exercise-Instruction.txt' in the 'ingress-exercise' repository. At the top, there's a branch selector set to 'main' and the file path 'ingress-exercise / Exercise-Instruction.txt'. Below this, a commit by 'leangsengk90' is shown with the message 'Update Exercise-Instruction.txt'. It indicates '1 contributor'. The file statistics show '13 lines (11 sloc)' and '546 Bytes'. The content of the file is a list of instructions for the exercise, numbered 1 through 13.

```
1 Exercise Instrucion:
2
3 1. Create Pipeline to deploy Wordpress with Mysql Services.
4   a. Deploy with Specific nodes.
5   b. Use Secret for all Environments in Mysql Deployment.
6   c. Use Configmap for all Environments in Wordpress Deployment.
7   d. Use NFS for Mysql Deployment
8   e. Use PV and PVC for Wordpress
9   f. Copy Yaml files from Jenkins server to Kubernetes server
10  g. Run those Yaml files in Kubernetes server
11
12 2. Create Ingress to expose Wordpress service and Mysql service
13 3. Apply TLS for those services with Cert-Manager + Let's Encrypt
```

A large, faint watermark of the ISTAD logo is centered in the background. It is a circular emblem with a blue outer ring containing the text 'INSTITUTE OF SCIENCE AND TECHNOLOGY ADVANCED DEVELOPMENT' and Khmer text. Inside the ring is a blue circle with a white atomic symbol. The word 'ISTAD' is written in red across the bottom of the inner circle.

Thank you

Begin with the theory, Start with the practical